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RESEARCH ARTICLE

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Economic evaluation of 22 HPV vaccination strategies to minimize the risk of penile, anal, and oropharyngeal cancers in Chinese males: A Markov model analysis

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ABSTRACT

Males can benefit significantly from human papillomavirus (HPV) vaccination. This study developed a decision-Markov model to evaluate the lifetime effectiveness and cost-effectiveness of 22 HPV vaccination strategies for preventing penile, anal, and oropharyngeal cancers. From a healthcare system perspective, vaccination strategies targeting males aged 14 and 40 were compared against no vaccination. Key parameters, including disease progression probabilities, cancer incidence, health state utilities, and direct medical costs, were sourced from the published literature and national databases. We calculated the number of cancer cases and deaths under different strategies and compared their cost-effectiveness. Results demonstrated that male vaccination is highly cost-effective. Vaccinating 14-y-old males averted between 73,724 and 416,654 cancer cases and between 1,102 and 6,229 deaths, depending on the vaccine valency and coverage rate. Similarly, strategies targeting 40-y-old males demonstrated substantial benefits. Overall, 20 of the 22 strategies were deemed highly cost-effective against a willingness-to-pay threshold of three times the national gross domestic product per capita. Sensitivity analysis confirmed the results' robustness. This study provides economic evidence to inform national HPV vaccination policies, demonstrating that including males, particularly adolescents, in vaccination programs represents a cost-effective investment for reducing the burden of HPV-associated cancers in China.

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Background

Human papillomavirus (HPV) is one of the most common sexually transmitted pathogens worldwide, and persistent infection can cause cervical, penile, oropharyngeal and other head and neck cancers. The study showed that 31% of males aged 15 y and above infected with at least one genital HPV and 21% infected with at least one high-risk or carcinogenic HPV.[1] The prevalence of HPV infection among Chinese men who have sex with men (MSM) ranges from 40% to 60%,[2] and the prevalence of HPV infection among HIV-positive males is as high as 33%–94.8%.[3] Approximately, 70,000 males suffer from cancer annually because of HPV infection worldwide.[4] More than 60% of penile cancers and 90% of anal cancers in men in the United States are caused by HPV.[5] The number of oropharyngeal cancers caused by HPV in males is also rising.[6,7]

HPV vaccination is considered to be an effective and cost-effective method of preventing HPV infection. It provides direct protection against genital warts and cancers (including penile, anal and oropharyngeal cancers) in males, reducing the risk of disease, as well as indirect protection and HPV prevention in females by reducing HPV transmission from infected males. In the UK, HPV vaccination of school-age boys yields

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significant clinical and economic benefits[8]; studies of HPV vaccine strategies in Japan and the USA have shown that HPV vaccination of males under a certain age is cost-effective,[9,10] while studies in the Sweden, and Norway show that HPV vaccination for males is cost-effective when the price is appropriate.[11,12]

The WHO recommends HPV vaccination for males, including adults and MSM, when feasible and affordable.[13] By 2023, 140 countries had incorporated HPV into their national programs, but only 47 extended vaccination to males.[14] Australia provided free 9-valent HPV vaccination for male adolescents aged 12–13 y in 2013[15]; South Korea provides free HPV vaccine for children[16]; and the UK not only provides free 9-valent HPV vaccination for male adolescents but also for MSM under the age of 45.[17,18]

In 2019, only 4% of males received HPV vaccination.[19] While some high-income countries have incorporated males into their HPV immunization programs, low- and middle-income countries have paid insufficient attention to male vaccination.[20] In China, a 4-valent HPV vaccine was approved in January 2025 for males aged 9–26 y, making the first HPV vaccine approved for males in the country. To date, there have been few studies on HPV vaccination among males in China, and economic evidence is lacking. This study aimed to construct a decision-Markov model to evaluate the lifetime effectiveness and economics of 22 HPV vaccination strategies for the prevention of penile, anal and oropharyngeal cancers and to provide evidence for the development of HPV vaccination programs for Chinese males.

Methods

From a healthcare system perspective, we constructed a decision-Markov model to compare the health outcomes and costs between HPV vaccination against no vaccination in males aged 14 and 40 y. Given that vaccination age, vaccine type and vaccine coverage would influence health outcomes, we developed different cohort models to analyze the cost and health utility of vaccines under different scenarios. Markov models can effectively simulate the long-term progression of HPV-related diseases, allow for flexibly specifying health states and transition probabilities, and support robust cost-effectiveness comparisons across vaccination strategies. We developed the Markov model using Treeage Pro 2022. Based on China's life expectancy of 77.93 y in 2020, the model time horizons were set to 64 y to capture the lifetime benefits of vaccine at age 14 and 30 y to approximate the health benefits of vaccination at age 40. To maintain data consistency, the parameters used in the model were adjusted to 2020 in accordance with the average exchange rate of the Chinese Yuan (CNY) against the US dollar and the Consumer Price Index. The discount rate was set to 3% as recommended by the WHO Health Economics Guidelines, to eliminate the impact of the time value of money.

Model and basic assumptions

The model contained 12 mutually exclusive health states representing different health conditions: healthy (with/without the vaccine), infected (with/without the vaccine), local cancer, regional cancer, distant cancer, and death (Figure 1). A model cycle-length of 1-y, with a half-cycle correction was assumed. Each person can transition or develop between different states each year, and individuals can remain in one state, progress to the next state, or die. Within each cycle, the patients are assumed to have only one type of cancer. The model also assumes that the cancer progression begins with localized cancer, progresses to regional cancer, and eventually develops into distant cancer.

Target population and disease

This study selected males aged 14 and 40 y in China as the target population. This was done for two reasons: 1) the vaccine achieves the best immune response before the first sexual intercourse, therefore, males aged 14 y were included[21]; and 2) the incidence of penile, anal and oropharyngeal cancers peaks between 40–70 y of age.[22] Therefore, males aged 40 were also selected. 4-valent and 9-valent HPV vaccines are available for males, and 2-valent vaccine was excluded from this model.

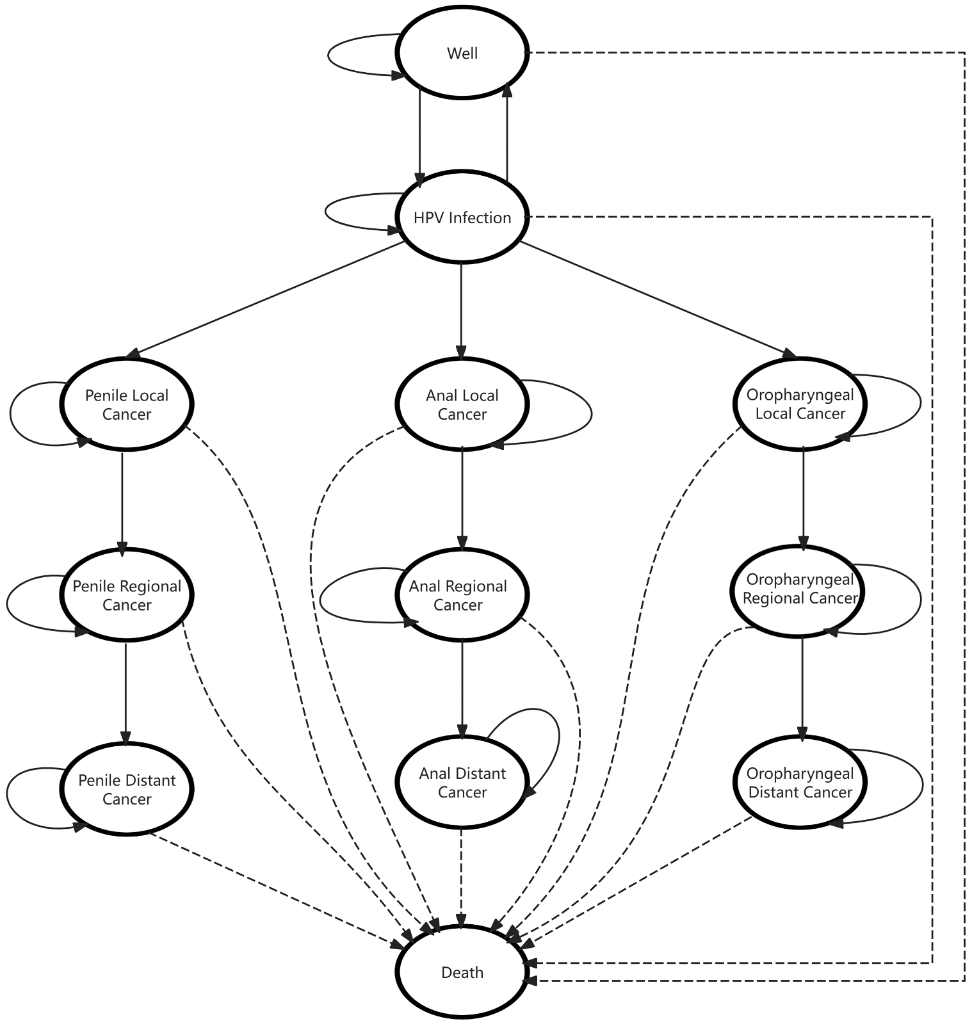

Figure 1. The health states used in the Markov model on HPV vaccination among males aged 14 or males aged 40. Circles represent health states. Arrows between circles indicate transitions between health states, while arrows that return to the same circle indicate remaining in the same state. Solid lines represent health state transitions, dotted lines represent death.

Vaccination strategies

Twenty-two HPV vaccination strategies were evaluated, including two target populations (males aged 14 and 40 y), vaccine coverage (10%, 20%, 30%, 40%, and 50%), and HPV vaccine type (4/9-valent HPV vaccine). (Details of the vaccination strategies appear in Supplementary Table s1). In accordance with China's current immunization guidelines and Leave no one behind: guidance for planning and implementing catch-up vaccination,[21,22] this study models set two doses of HPV vaccine for 14-y-old boys and three doses for 40-y-old men.

Model inputs

The model only considered direct costs including HPV vaccine costs, outpatient costs, and treatment costs (costs of treating non-cancer diseases caused by HPV infection, and costs of treating penile, anal and oropharyngeal cancers). Outpatient costs were applied only to individuals infected with HPV, because HPV screening for the target population was not included in the analysis. Healthy individuals received 1 quality-adjust life year (QALY) per cycle. HPV-infected individual loses 0.04 QALY per cycle; therefore, the QALY for the infection was 0.96. The QALY for patients with penile, anal and oropharyngeal cancers were 0.79, 0.57, and 0.58, respectively. The QALY for

Table 1. Parameter values and data sources of MARKOV model.

Parameters	Value (Range)	Resources
Costs (2020 ¥)		
4-valent vaccine per dose	850.00(680.00 ~ 1020.00)	23
9-valent vaccine per dose	1380.00(1104.00 ~ 1656.00)	23
HPV outpatient	316.53(253.22 ~ 379.83)	24
HPV treatment	0.00(0.00 ~ 1000.00)	
PCl treatment	45418.56(36334.84 ~ 54502.27)	25
PCr treatment	66998.82(53599.05 ~ 80398.58)	25
PCd treatment	85088.83(68071.06 ~ 102106.59)	25
ACl treatment	50618.46(40494.77 ~ 60742.16)	25
ACr treatment	66605.28(53284.22 ~ 79926.33)	25
ACd treatment	94875.75(75900.6 ~ 113850.91)	25
OCl treatment	42863.76(34291.00 ~ 51436.51)	25
OCr treatment	59276.38(47421.10 ~ 71131.65)	25
OCd cancer treatment	78463.14(62770.51 ~ 94155.76)	25
Probabilities	(%)	
HPV infection	14.50(11.30 ~ 17.70)	26
Recovery from HPV infection	58.00(53.00 ~ 63.00)	27
PC incidence	0.0041(0.0033 ~ 0.0050)	28,29
AC incidence	0.0049(0.0039 ~ 0.0058)	28,29
OC incidence	0.0063(0.0050 ~ 0.0075)	28,29
Transition probability		
PCl to PCr	21.00(16.80 ~ 25.20)	25
PCr to PCd	54.70(43.76 ~ 65.64)	25
PCl to death	22.30(17.84 ~ 26.76)	25
PCr to death	45.30(36.24 ~ 54.36)	25
PCd to death	58.80(47.04 ~ 70.56)	25
ACl to ACr	23.90(19.12 ~ 28.68)	25
ACr to ACd	50.00(40.00 ~ 60.00)	25
ACl to death	18.10(14.48 ~ 21.72)	25
ACr to death	33.00(26.40 ~ 39.60)	25
ACd to death	50.10(40.08 ~ 60.12)	25
OCl to OCr	23.90(19.12 ~ 28.68)	25
OCr to OCd	54.70(43.76 ~ 65.64)	25
OCl to death	16.30(13.04 ~ 19.56)	25
OCr to death	16.30(13.04 ~ 19.56)	25
OCd to death	40.80(32.64 ~ 48.96)	25
Utility		
Healthy	1.00	
HPV infection	0.96	30
PC	0.79(0.74 ~ 0.84)	31
AC	0.57(0.52 ~ 0.62)	31
OC	0.58(0.53 ~ 0.63)	31
Others	(%)	
Discount rate	3.00	WHO Guide on Standardization of Economic of Evaluations of Immunization Programmes
4-valent vaccine efficacy	0.90(0.85 ~ 0.95)	32
9-valent vaccine efficacy	0.94(0.89 ~ 0.99)	33

PC: penile cancer; AC: anal cancer; OC: oropharynx cancer; l: localized; r: regional; d: distant.

the death was 0. Individuals undergoing state transitions could occur uniformly over the cycle, either at the beginning or the end. Therefore, the health utility of those who experienced transition was calculated as the average of the health utility values of the initial and final states.

In 2020, there were 8,377,653 males aged 14 y and 9,785,394 males aged 40 y in China. Parameters including the probability of HPV infection, cancer incidence and mortality, QALY, vaccine costs, and treatment costs were obtained from the literature and national/public data (Table 1).

Health outcomes

This study focused on two health outcomes: the lifetime benefits of HPV vaccination among males aged 14, and the health benefits gained after vaccination for males aged 40. The model also estimated the vaccination costs across different ages, vaccine types, and coverage levels, as well as the cumulative number of HPV-related cancer cases and deaths under various vaccination strategies.

Economic evaluation

The incremental cost-utility ratio (ICUR) was used to identify the most cost-effective strategy. A strategy was considered highly cost-effective if the ICUR was less than one time the gross domestic product (GDP) per capita, cost-effective if the ICUR ranged between one and three times the GDP per capita, and not cost-effective if the ICUR exceeded three times the GDP per capita. In 2020, China's GDP per capita was ¥72,447.

Sensitivity and uncertainty analyses

Univariate and probabilistic sensitivity analyses were conducted to assess the results' robustness. In the univariate sensitivity analyses, parameters were varied within the following ranges: cost parameters (±20% or 95% uncertainty interval [UI]), HPV infection probability (95% UI), HPV transition rate (±20%), and vaccine efficacy (±5%). The results were presented using tornado plots. For the probabilistic sensitivity analysis, Monte Carlo simulations were performed with 1,000 iterations and parameters from their respective probability distributions to calculate the costs, utility, and ICUR for the optimal vaccination strategy. Cost parameters were assumed to follow Gamma distributions, whereas the probability parameters followed Beta distributions. The results were visualized using an acceptability curve.

Results

Costs

Males aged 14 and 40 would incur treatment costs of ¥10.95 billion and ¥8.69 billion without vaccination, respectively. Using 50% vaccine coverage scenarios (Strategy 10, 11, 21, and 22) as an example, the total costs of vaccination for males aged 14 with the 4-valent and 9-valent vaccines would be ¥14.54 billion and ¥17.93 billion, respectively. Of these, vaccine costs accounted for ¥7.95 billion and ¥11.56 billion, respectively, and the treatment costs accounted for ¥6.59 billion and ¥6.37 billion, respectively. For males aged 40, the total costs of vaccination with the 4-valent and 9-valent vaccines would be ¥19.16 billion and ¥25.3 billion. This included ¥13.94 billion and ¥20.26 billion vaccine costs, and ¥5.21 billion and ¥5.04 billion treatment costs, respectively. The costs of the 22 strategies are listed in Table 2.

Health outcomes

Using 50% vaccine coverage as an example (Strategy 10, 11, 21 and 22), compared against no vaccination, vaccination for males aged 14 y with 4-valent and 9-valent vaccines would reduce 397,205 and 416,654 HPV cases, and avert 5,939 and 6,230 deaths, respectively, and vaccination for males aged 40 y with 4-valent and 9-valent vaccines would reduce 620,357 and 650,764 cases, and 3,909 to 4,099 deaths, respectively (Figure 2).

At vaccine coverage ranging from 10% to 50%, vaccinating males would reduce the number of cases by 7.36% to 41.60% and deaths would be reduced by 0.03% to 0.19%. Similarly, vaccinating men aged 40 y would reduce the number of cases by 7.34% to 41.54% and deaths would be reduced by 0.03% to 0.22%. The numbers of cases and deaths averted under the different strategies for HPV infection are shown in Supplementary Table s2.

Cost-effectiveness analyses

Using 50% vaccine coverage as an example (Strategy 10, 11, 21 and 22), vaccinating males with 4-valent and 9-vanlent vaccines incurred additional costs of ¥3.60 billion and ¥6.99 billion, respectively. The cost per case averted was ¥9,056.70 and ¥16,768.93, and cost per death averted was ¥605,721.22 and ¥1,121,543.16, respectively. For males aged 40 y with 4-valent and 9-vanlent vaccines resulted in additional cost of ¥10.47 billion and ¥16.61 billion, respectively. The cost per case averted was ¥16,877.28 and ¥25,527.48, and cost per death averted was ¥2,678,640.75 and ¥4,052,479.80, respectively. The number of cases and deaths averted and the cost-effectiveness of the 22 strategies are shown in Supplementary Table s3.

Table 2. Vaccination costs, treatment costs, and total cost under 22 different vaccination strategies.

Strategy	Vaccine cost (CNY billion)	Treatment cost (CNY billion)	Total cost (CNY billion)
Strategy 1	0.00	10.95	10.95
Strategy 2	1.59	10.14	11.73
Strategy 3	2.31	10.10	12.41
Strategy 4	3.18	9.30	12.48
Strategy 5	4.62	9.22	13.84
Strategy 6	4.78	8.42	13.20
Strategy 7	6.94	8.31	15.24
Strategy 8	6.37	7.52	13.89
Strategy 9	9.25	7.33	16.58
Strategy 10	7.95	6.59	14.54
Strategy 11	11.56	6.37	17.93
Strategy 12	0.00	8.69	8.69
Strategy 13	2.79	8.04	10.83
Strategy 14	4.05	8.01	12.06
Strategy 15	5.58	7.37	12.95
Strategy 16	8.10	7.31	15.41
Strategy 17	8.37	6.68	15.04
Strategy 18	12.15	6.58	18.73
Strategy 19	11.16	5.96	17.11
Strategy 20	16.20	5.81	22.01
Strategy 21	13.94	5.21	19.16
Strategy 22	20.26	5.04	25.30

Strategy 1–11 apply to 14-y-old males: Strategy 1 represents no HPV vaccination, while Strategies 2–11 are vaccination with 4-valent or 9-valent vaccination at coverage of 10%, 20%, 30%, 40%, or 50%, respectively. Strategy 12–22 apply to 40-y-old males. Strategy 12 is no HPV vaccination; Strategy 13–22 are vaccination with 4-valent or 9-valent vaccination at coverage of 10%, 20%, 30%, 40%, or 50%, respectively.

Cost-utility analyses

Using 50% vaccine coverage as an example (Strategy 10, 11, 21 and 22), total utility of males aged 14 y with 4-valent and 9-vanlent vaccines were 1.78 million QALY and 1.86 million QALY, with ICUR of ¥2024.78 per QALY and ¥3749.14 per QALY; similarly, total utility of males aged 40 y with 4-valent and 9-vanlent vaccines were 0.89 million QALY and 0.94 million QALY, with ICUR of ¥11,729.70 per QALY and ¥17,745.76 per QALY (Table 3).

All vaccination strategies across the ages groups, vaccine types, and vaccine coverage fell below the highly cost-effective willingness-to-pay (WTP) threshold. Although the 9-valent vaccine generated greater health benefits for males, its higher cost resulted in a less favorable ICUR than the 4-valent vaccine at the same coverage levels. Among the 22 vaccination strategies evaluated, the optimal vaccination strategies was 14 y old males to be vaccinated with 4-valent HPV vaccine at 50% coverage.

Sensitivity analyses

Across all parameter variations, the ICUR remained below 1-times GDP per capita across all parameter variations, demonstrating the results' robustness (Supplementary Figure s1). Under the optimal vaccination strategy by different age, the principal determinants of cost-utility were treatment costs, vaccine costs, and the probability of HPV infection.

As Figure 3 illustrated, among males aged 14 y, Strategy 11 demonstrated the highest acceptability when the WTP threshold exceeded ¥40,000. Among males aged 40 y, strategy 12 was more acceptable than other vaccination strategies when the WTP threshold was below ¥70,000; however, Strategy 22 became more favorable at higher WTP thresholds.

Discussion

We evaluated HPV vaccination's cost and benefits for males in China. HPV vaccination for males aged 14 y averted 73,724 (4-valent HPV vaccine, 10% coverage) to 416,654 (9-valent HPV vaccine, coverage 50%) cases and 1,102 to 6,229 deaths. HPV vaccination for males aged 40 y averted 115,048 (4-valent

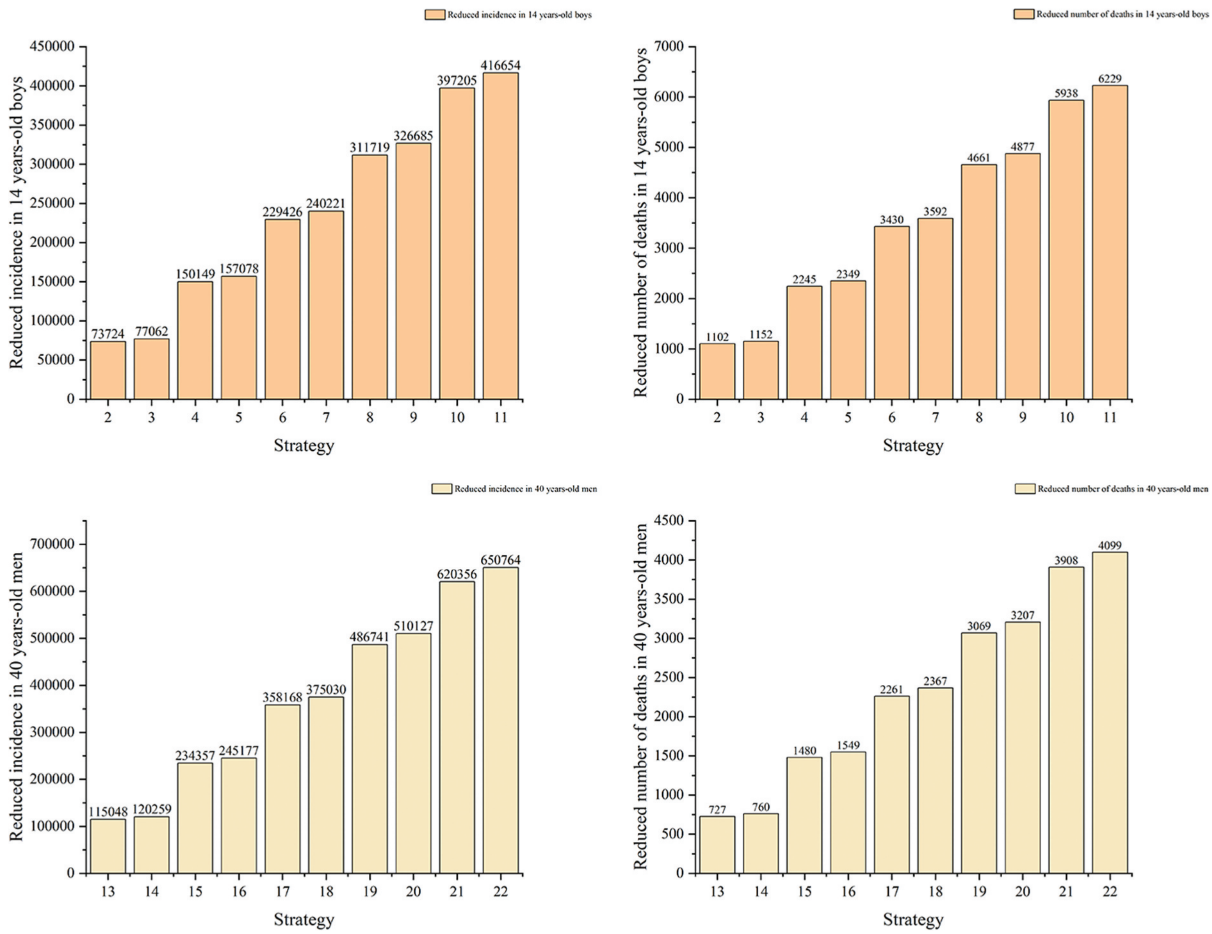

Figure 2. Number of cases and deaths averted by vaccination in males under 20 different strategies. Strategy 2–11 indicate that males aged 14 y receive the 4-valent and 9-valent vaccine at coverage of 10%, 20%, 30%, 40%, and 50%, respectively. Strategy 13-22 indicates that males aged 40 y receive the 4-valent and 9-valent vaccine at coverage of 10%, 20%, 30%, 40%, and 50%, respectively.

HPV vaccine, 10% coverage) to 650,764 (9-valent HPV vaccine, coverage 50%) cases and 727 to 4,099 deaths. Although the results indicate that HPV vaccination prevents more cases among 40-y-old males than among 14-y-olds, this difference is primarily due to variation in the underlying population size (9.785 vs. 8.378 million in China) and the later peak in cancer risk (oropharyngeal, penile, and anal cancers peak after age 45 in Chinese men). As a result, mass vaccination at the age of 40 y could prevent a greater absolute number of cases. However, assuming equivalent coverage and efficacy, vaccinating at 14 y old is more cost-effective for disease prevention because it confers earlier protection and longer-lasting immunity.

The results demonstrated that 20 vaccination strategies were cost-effective at 3 times the GDP per capita, consistent with findings from the USA,[34] Netherlands [35] and Singapore.[36] Among these strategies, vaccinating 50% of males aged 14 y emerged as the optimal approach, aligning with cost-effectiveness evaluations of HPV vaccination in females across different age groups, which consistently show that adolescent vaccination is more cost-effective than vaccination in adulthood.[37] Current efforts to prevent and control HPV-related diseases in China primarily focus on women, and greater attention should also be paid to male vaccination. Additional measures are needed to enhance vaccination impacts, including strengthened health education and promotion,[38] and reduced vaccine price to improve vaccine coverage.[39]

Regarding vaccine types, the efficacies of 4-valent and 9-valent vaccines were about 90% and 95%. Although the 9-valent vaccine could achieve higher utility, the 4-valent vaccine demonstrated a notable economic advantage in this study, which because of its lower price. From an economic perspective, the

Table 3. Baseline cost-utility analysis under 22 vaccination strategies.

Strategy	Total cost (CNY billion)	QALY (million)	Incremental cost (CNY billion)	Incremental QALY (CNY million)	ICUR (CNY per QALY)
Strategy 1	10.95	492.96	—	—	—
Strategy 2	11.73	493.29	0.78	0.33	2364.97
Strategy 3	12.41	493.30	1.46	0.34	4244.97
Strategy 4	12.48	493.63	1.53	0.67	2279.87
Strategy 5	13.84	493.66	2.90	0.70	4120.95
Strategy 6	13.20	493.99	2.25	1.03	2194.80
Strategy 7	15.24	494.03	4.30	1.07	4003.74
Strategy 8	13.89	494.35	2.94	1.39	2109.77
Strategy 9	16.58	494.42	5.63	1.46	3852.48
Strategy 10	14.54	494.74	3.60	1.78	2024.78
Strategy 11	17.93	494.82	6.99	1.86	3749.14
Strategy 12	8.69	274.04	—	—	—
Strategy 13	10.83	274.21	2.14	0.17	12884.30
Strategy 14	12.06	274.22	3.37	0.17	19425.01
Strategy 15	12.95	274.38	4.26	0.34	12595.54
Strategy 16	15.41	274.40	6.72	0.35	19005.05
Strategy 17	15.04	274.56	6.36	0.52	12306.85
Strategy 18	18.73	274.58	10.05	0.54	18617.20
Strategy 19	17.11	274.74	8.43	0.70	12018.23
Strategy 20	22.01	274.78	13.32	0.73	18135.35
Strategy 21	19.16	274.94	10.47	0.89	11729.70
Strategy 22	25.30	274.98	16.61	0.94	17745.76

Incremental costs, incremental QALY, and ICUR are the increased costs, QALY, and incremental cost-utility ratios at the same age of vaccination compared against no vaccination; total costs include the cost of vaccination and treatment.

Strategy 1–11 target 14-y-old males: Strategy 1 represents no HPV vaccination, while Strategy 2–11 correspond to vaccination coverage rate of 10%, 20%, 30%, 40%, and 50% for both 4-valent and 9-valent vaccines. Strategy 12–22 target 40-y-old males: Strategy 12 represents no HPV vaccination, while Strategies 13–22 correspond to vaccination with the 4-valent and 9-valent vaccines at coverage rate of 10%, 20%, 30%, 40%, and 50%, respectively.

4-valent vaccine offers a clearer cost advantage than 9-valent vaccine, highlighting the need to balance vaccine cost with demand. Promoting diversified financing mechanisms to reduce vaccine prices could deliver more effective health protection under limited resource conditions.

Compared with the vaccine economic evaluations focused on a single disease, HPV vaccination delivers a broad range of health benefits. Vaccination can prevent over 90% of cancers caused by HPV, as well as anal, vaginal, cervical, and vulvar precancerous lesions. Accordingly, this study considered vaccine effectiveness in preventing three cancers, yielding estimates of health gains that exceed those reported in studies limited to selected diseases.

In 2025, China took a significant step forward by integrating the HPV vaccine into its national immunization program, offering free vaccination to 13-y-old girls. Based on the findings of this study, China should further expand HPV vaccination coverage by prioritizing the inclusion of adolescent males in the program, which would promote immunization equity and contribute to the development of a more comprehensive disease prevention system. At the same time, the range of vaccines available to males should be expanded by accelerating the introduction of domestically produced vaccines suitable for male recipients, increasing vaccine supply, and lowering the economic barriers to vaccination, thereby enhancing accessibility and equity. Furthermore, integrated strategies should be implemented to increase vaccination coverage, combining sustained public education efforts with the exploration of feasible cost-subsidy mechanisms.

This study had some limitations. First, the model focused on preventing HPV-related cancers and did not account for protection against other HPV-related outcomes, such as genital warts, or herd immunity from vaccination. As a result, the health benefits may be underestimated and may overestimate the ICUR. Second, the model assumed that all target individuals are HPV-free at enrollment, whereas real cohorts may include pre-infected individuals with lower health utility. This may overestimate QALY gains and underestimate the ICUR. Third, it assumed lifelong post-vaccination protection and does not account for potential booster doses, which could lead to underestimating ICUR. Fourth, the model did not include possible risk-compensation behaviors after vaccination (e.g., reduced screening) or variations in real-world vaccination adherence, both of which may affect disease prevention and cost-effectiveness estimates. Finally, despite sensitivity analyses indicated robust results, uncertainty remains in several parameters including HPV progression probabilities and long-term real-world vaccine effectiveness, which were derived from published and macro-level data. Future research should address these uncertainties as higher-

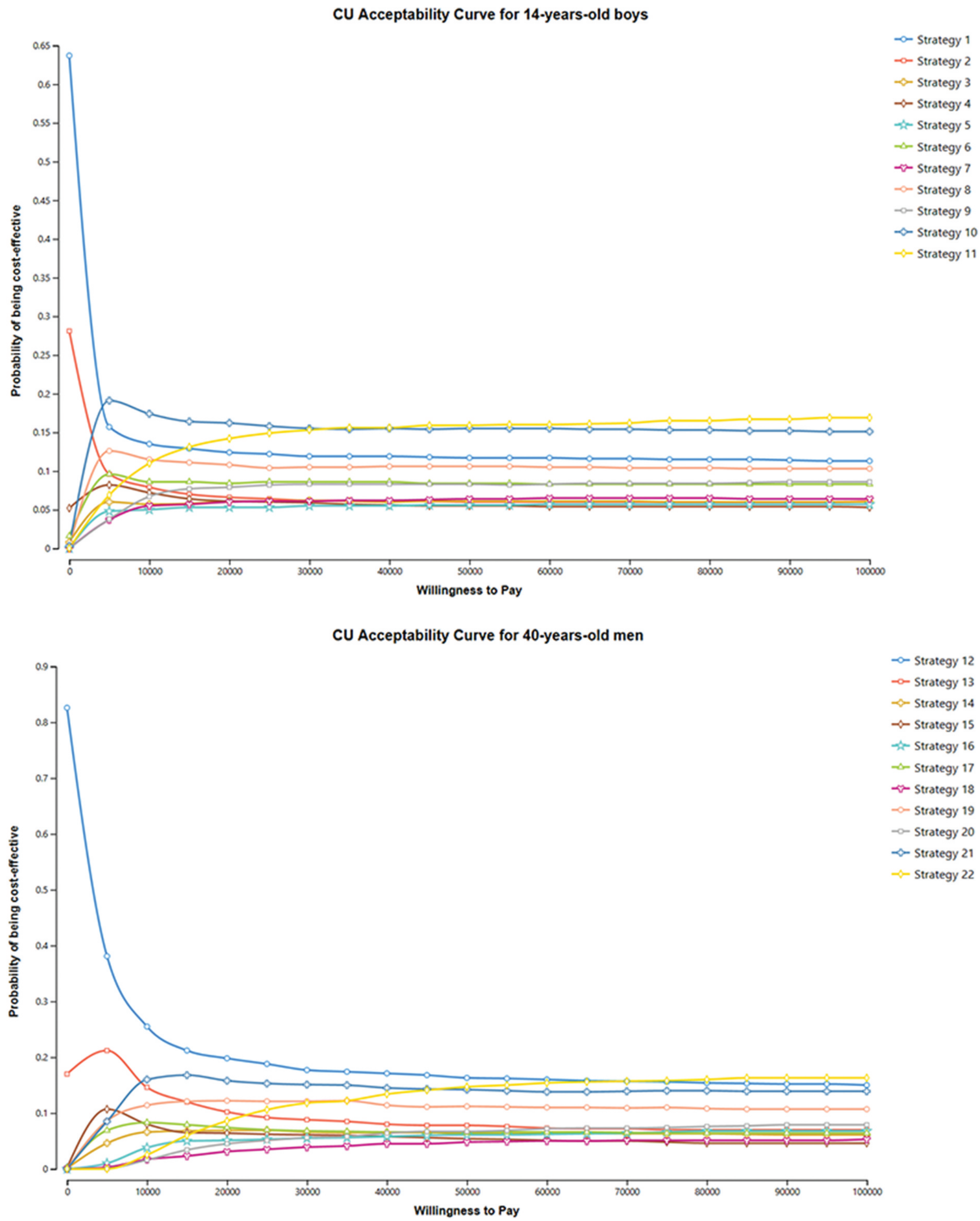

Figure 3. Cost-utility acceptability curve for males aged 14 and 40 under 22 different strategies. The top panel shows the probability of accepting each vaccination strategy for males aged 14 across different WTP thresholds. The bottom panel shows the corresponding probabilities of acceptance for males aged 40 across the same range of WTP levels.

quality local data become available and should incorporate dynamic transmission models with real-world evidence to better capture herd immunity and evaluate the long-term effects of complex vaccination strategies.

Conclusion

This study evaluated the cost-effectiveness of 22 HPV vaccination strategies for Chinese males in preventing three HPV-related cancers using a decision-analytic Markov model. The results confirm that HPV vaccination for males is cost-effective, with the 4-valent vaccination of 14-y-old boys identified as the most favorable strategy. Our findings provide important economic evidence to support the inclusion of males, particularly adolescents in China's national immunization program.

Author contributions

CRediT: **Di Wang:** Conceptualization, Formal analysis, Writing – original draft; **Weihua Luo:** Conceptualization, Writing – original draft; **Ruixi Qin:** Data curation; **Gan Xu:** Writing – review & editing; **Peipei Chai:** Writing – review & editing; **Bingjie Liu:** Data curation; **Liangru Zhou:** Conceptualization, Writing – review & editing; **Xin Zhang:** Writing – review & editing.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Abbreviations

GDP	Gross domestic product
HPV	Human papillomavirus
ICUR	Incremental cost-utility ratio
MSM	Men who have sex with men
QALY	Quality-adjusted life years
UI	Uncertainty interval
WTP	Willingness-to-pay

Data availability statement

The data of this study are available from the authors.

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